

Pratyush Bindal

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Education

Birla Institute of Technology and Science, Pilani

Aug 2022 – July 2026

Bachelor of Engineering in Computer Science, Minor in Data Science (CGPA: 9.61 / 10.00)

Hyderabad, Telangana

- **Relevant Coursework:** Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Database Systems, Object Oriented Programming, Operating Systems, Computer Architecture, Computer Networks, Compilers

G. D. Goenka Public School

April 2018 - July 2022

High School (Class XII: 96.6% and Class X: 97.6%)

Muzaffarnagar, Uttar Pradesh

Experience

Intern, Member of Technical Staff

Jan 2026 – Present

Nutanix Technologies

Bengaluru, Karnataka

- Migrated critical production workflows from a legacy microservice architecture to a Go-based distributed worker-executor framework, improving scalability, fault isolation, and execution reliability

Software Engineer Intern

May 2025 – July 2025

Microsoft Corporation

Hyderabad, Telangana

- Performed startup analysis on Android Office Mobile apps using profiling tools to identify bottlenecks, and executed critical optimizations that reduced latency and significantly improved responsiveness at scale

Summer Research Intern

May 2024 – July 2024

Indian Institute of Remote Sensing (IIRS), Indian Space Research Organization (ISRO)

Dehradun, Uttarakhand

- Led a 4-member team to develop advanced ship-detection models, including CNN-based models like YOLO v8, v9, v10 and DETR-based techniques like Grounding DINO for accurate maritime vessel recognition in satellite images

Skills

Languages: C, C++, Java, Python, Golang, Kotlin, SQL, HTML, CSS, JavaScript, TypeScript

Frameworks, Libraries & Tools: TensorFlow, PyTorch, Spring Framework, Android Studio, Git

Databases: MySQL, Supabase, MongoDB, Firebase Realtime DB

Projects

DRL-based Anomaly Detection in WBANs | Machine Learning

- Built a DRL-based anomaly detection system for healthcare WBANs, with an end-to-end web application and LLM-powered query interface (**Patent Application No.: 202511091698**)

Intrusion Detection System | Deep Reinforcement Learning

- Implemented AE-SAC and TD3 reinforcement learning models on the NSL-KDD dataset to develop an efficient intrusion detection system for improved accuracy and robustness in cybersecurity

Multi-Elevators Scheduling System | Operating Systems

- Built a multi-threaded elevator scheduling system in C using POSIX threads, shared memory, and message queues, with lock-free request handling and parallel scheduling to support 100 elevators across 500 floors efficiently

Q-Quest | Reinforcement Learning

- Implemented and benchmarked Q, Double-Q, Triple-Q, and Quadruple-Q Learning algorithms in a custom RL environment, demonstrating superior stability and bias reduction of Double-Q Learning

Toy Compiler Development | Compiler Construction

- Developed a compiler from scratch using Flex/Bison, enforcing lexical analysis, syntax parsing, AST generation, and three-address code for expression evaluation and control flow

Co-Curriculars

- Teaching Assistant: BITS F464 Machine Learning (Semester I & II, 2025–26) & Database Systems (WILP Course, 2025)
- Undergraduate Mentor: MATH F111 Mathematics I (Semester II, 2022–23)

Honours and Awards

- Awarded 100% Merit Scholarship (top 1%) and 80% Merit-cum-Need Scholarship (top 3%) for all semesters